Nearly right, and still too sure of itself

A registry has the coverage and the confounding; a trial has the identification and no reach. Folding one into the other is the obvious move. The question is which of the two available ways to do it, and what each one leaves you believing.

Abstract

Objective. A registry has the coverage and the confounding; a trial has the identification and no reach. Folding one into the other is the obvious move. The question is which of the two available ways to do it, and what each one leaves you believing. **Methods.** Build the registry, including the severity nobody recorded. 6 step(s) are recorded in full below. **Results.** The uncalibrated registry was not merely uncertain, it was confidently wrong: amplitude 14.89 against a truth of 8.0, which is 2.8 posterior standard deviations out, and it openly disagreed with the trial (z = -2.44). Tight and wrong is the signature of confounding rather than noise. **Conclusions.** No assumptions were recorded for this run, which is not the same as none being required.

1. Introduction

This report addresses one question: A registry has the coverage and the confounding; a trial has the identification and no reach. Folding one into the other is the obvious move. The question is which of the two available ways to do it, and what each one leaves you believing.

2. Methods

Question. A registry has the coverage and the confounding; a trial has the identification and no reach. Folding one into the other is the obvious move. The question is which of the two available ways to do it, and what each one leaves you believing.

Build the registry, including the severity nobody recorded

Confounding by indication is the defining problem of registry evidence: sicker patients are dosed harder, so the raw association understates the benefit. Simulating it explicitly — with a severity term that enters both the dose and the outcome and is never handed to the model — is what makes the rest of this checkable, because the true amplitude is known.

considered instead: Simulating clean data and calling the result a demonstration of calibration. Calibration on unconfounded data does nothing interesting; the whole operation exists to correct a bias, so the bias has to be there.; readout: registry : 6 clinics x 24 periods; true amplitude : 8.0; correlation(dose, unmeasured severity) = 0.481; The severity column exists in this script and is never passed to fit().

Say precisely what the trial measured, as an object

A trial result is not a number, it is a number attached to a question: which contrast, over which window, in which population, at which level. The Estimand records all of that and hashes it, so a measurement can only be folded into a model that answers the same question. The hash is what stops a first-period per-patient contrast being quietly used to constrain a cumulative population-level one.

considered instead: Carrying the trial as 'the treatment effect was 12.4'. That is the form in which evidence is normally transferred between studies, and it is the form in which the window, the population and the estimand silently change on the way.; readout: trial estimate : 5.215 +/- 0.193; interval : [4.83679, 5.59274] (95% WALD); n : 280 patients, 1 period; target hash : 2949465ce0427194...

Fit the registry alone, and notice it is not merely uncertain

The failure mode worth recognising is tight and wrong. If the uncalibrated fit were simply noisy, its interval would cover the truth and the trial would be a precision upgrade. It does not: the truth sits several posterior standard deviations away, which is the signature of confounding rather than sampling error, and no amount of registry data would fix it.

considered instead: Judging the fit by its convergence diagnostics and its posterior predictive check. Both pass here. A confounded model fits its own data beautifully — that is what makes confounding dangerous rather than obvious.; readout: converged : True; amplitude : 14.891 +/- 2.428; truth : 8.0; drift vs truth : +2.8 posterior sd

Route one: turn the trial into a prior, then refit

derive_prior converts the trial's statement about a PREDICTION into a statement about a PARAMETER, using a design factor that carries the conversion. Every step of that conversion writes a ledger line, so the assumption that made it possible is recoverable later — this is the transfer that most often disappears into a footnote.

considered instead: Setting an informative prior by hand to match the trial. It produces a similar posterior and records nothing: six months later there is no way to tell which prior came from evidence and which came from an analyst's judgement about what the answer ought to be.; readout: design factor : 0.5478; parameter : beta_dose; new prior : family='lognormal' hyper={'mu': 2.2526030052579893, 'sigma': 0.03696835710791612}; ledger:

Route two: add the trial to the log density instead

fit_calibrated puts the measurement into the objective as a constraint, so the trial competes with the registry data rather than preceding it. When the two disagree the posterior lands between them, weighted by their precisions, instead of the registry being fitted inside a prior the trial already fixed.

considered instead: Assuming this is the same operation as route one with the arithmetic rearranged. It is not, and the two do not agree — constraining a parameter and constraining a prediction are different requests. Which you want depends on what the model is for.; readout: amplitude : 11.892 +/- 1.900

Compare all three on the two things that matter

Distance from the truth and agreement with the trial are different questions and a fit can pass one while failing the other. Reporting the drift in posterior standard deviations rather than in raw units is what exposes that: a fit can land closest in absolute terms and still be the worst on this column, because it left itself too little uncertainty to be wrong in.

considered instead: Picking a winner. axiom reports the gap between the two routes rather than choosing, because the choice depends on what the model will be used for — a formulary decision about this dose wants the constrained prediction, a model that will be extrapolated to other doses wants the constrained parameter.; readout: uncalibrated 14.891 2.428 drift +2.8 sd z vs trial -2.44 disagrees; prior route 9.733 0.349 drift +5.0 sd z vs trial -0.74 agrees; likelihood route 11.892 1.900 drift +2.0 sd z vs trial -0.55 agrees

3. Results

No findings were recorded.

4. What the run showed

Written by the analyst after seeing the output, and reproduced unchanged. Nothing in this section is generated.

The uncalibrated registry was not merely uncertain, it was confidently wrong: amplitude 14.89 against a truth of 8.0, which is 2.8 posterior standard deviations out, and it openly disagreed with the trial ($z = -2.44$). Tight and wrong is the signature of confounding rather than noise.

Both routes cut most of that bias and both then reconcile with the trial: 9.73 and 11.89 against an uncalibrated 14.89. Neither reaches the truth, and neither interval covers it. One trial of 280 patients does not undo confounding in a registry — it bounds it, and that is a smaller claim than the word 'calibrated' invites.

They also disagree with each other, by 2.16. That disagreement is the useful part rather than a defect to be averaged away: it is the size of the difference between the two questions being asked, and it is reported rather than resolved on your behalf.

The prior route lands CLOSEST in absolute terms and still scores worst on drift, because the derived prior inherited the trial's precision and left the posterior very tight: 0.349 against 1.900. A calibrated model can be nearly right and still overconfident. That is exactly the failure a recovery test against a known truth exists to catch, and the reason the design factor and the moment-matching wrote themselves into the ledger rather than disappearing into the fit.

5. Model checking

No diagnostics were recorded. An analysis with no model checking is not therefore sound; it is unexamined.

6. Discussion

There are no findings to interpret.

7. Conclusions

A registry has the coverage and the confounding; a trial has the identification and no reach. Folding one into the other is the obvious move. The question is which of the two available ways to do it, and what each one leaves you believing.

No finding was recorded, so no conclusion follows.

8. Limitations

No assumption in the record is left unresolved. That is a statement about the record, not a guarantee about the world.

9. Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Field	Value
field	Clinical research
source	examples/ walkthrough record
steps	6
evidence hash	4460c14079f43c654ebf3bdf7638749bb3d1598d73d189507def7db74df8c545

Table 1. Run